Glass Heroes Business Manager v0.3.3.6 — EXPERIMENTAL DEVELOPMENT REVIEW

Issues #33 (P0 service recovery) and #34 (business/UI/demand). Source-verified candidate, not a released RC or stable build. GTA/Windows execution NOT RUN; Enhanced 3.9.0.6 compatibility NOT VERIFIED. No merge, tag, release, installation, live save change or Issue28 crash attribution.

Evidence and baseline

Initial authorized feature branch feature/shvdn3-336-service-recovery-business-polish at 52749f7 contains only the handoff above approved .5 source commit 22bff0990bdc771215ebcb587e8426059df14336. Its C# SHA256 exactly matched 741c97730856adad11cd98d985e41d4eb325c9788af5ceebeee965a0eb64125f before edits. Read full committed CODEX_V0_3_3_6_OWNER_QA_HANDOFF.md and public Issues33/34. PR32/.5 and older candidate worktrees are protected, not merged or modified.

Original Errors log and owner Save.ini were NOT supplied in this workspace for .6. Embedded issue/handoff evidence reports: .5 GH+LS RED session without crash/managed exception; 16 saved-waypoint and 5 emergency walking recoveries, 3 ambiguous-direction mechanic parking failures, 14 queued parking failures, one direct arrival timeout, four normal departures and some repair-revenue entries. This is reported evidence, not a claim we read private full files or reproduced GTA.

Source confirms unset selected work nodes are skipped; mask8 selects point4, so the reported unset ServiceWaypoint4 produces no selected work stop. It does NOT independently explain pickup: pickup reverses ArrivalWaypoints, not work masks. Reported 17m/15m/16m/17m reverse pickup legs and Bay3 ~0.39m work/car separation are geometry hypotheses needing GTA checks. No saved path sorting, mask migration or coordinate rewrite occurs.

Source confirms paid and unpaid parking failures shared cancellation/message; Employee Status could count a stale working phase without a live repair. Those mechanisms are corrected/instrumented. Neither source tests nor the prior crash-free .5 session prove .6 fixes every navigation/animation failure or historical crash.

Focused commits and verification plan

P0 was implemented and tested before independent business features:

- 9889842: paid/unpaid direction handling, one parking retry, retained paid recovery, bounded driver exit, logical worker/return ownership, queue retry, mask warnings/USE-UNSET and initial accurate status. Initial gate 701/701 (683 inherited + 18 new), 21 static, actual Release 0 errors/two inherited warnings. Final evidence below supersedes initial source hashes.

- ae6dc03: owner amount/review/confirmation and removed direct cash shortcuts. Inherited finance33/startup18 plus new owner13.

- c5a9886: quantity retail draft/review/atomic batch, compatible audit and unchanged pricing. Original one-unit helper remains as a wrapper; inherited sale/repair suites plus new retail tests.

- e1f1339: employee snapshot menu, Settings.ini controller precedence, obsolete controller menu removal, row-specific help and input/navigation tests.

- 7db0764: GTA-hour-weighted active-time demand, new demand13.

- aca343b: NPC destination clearance and invalid-speed holds for paid recovery; normal OFF flags preserved.

- dcde623: negative unit-estimate guard and clearer workstation/demand labels.

- 8833abe: manual recovery restricted to physical handoffs; original OFF safe pickup remains automatic. Final recovery21.

- fc1d986: full staff summary uses the same refreshed/live-validated snapshot as the menu, eliminating stale receptionist status; status15.

- Final test/report commit: full evidence, fixture compatibility, 67-menu/392-footer matrix, source diff/hash and versioned DOCX/PDF; exact SHA in branch log/completion handoff.

No speculative new scheduler, placement engine, path reordering, background task or completed-car serialization. Each new recovery/transaction/demand condition has focused deterministic tests. Native engine behavior remains the owner acceptance gate.

Phase 1 — reliability and diagnostics

Routes, collection and workstation access

Five physical ServiceWaypoint positions remain SHARED geometry; JobOutboundMask/JobReturnMask remain per-mechanic selections. EffectiveWorkRouteMask intersects selected bits with saved flags, without modifying either. WorkRouteStatus shows selected/effective/unset masks; selected-but-unset rows show USE / UNSET. WarnEmptyWorkRoute emits one warning per slot/direction/mask combination in a bounded session set, only when building a route. A zero effective selection explicitly explains the inherited final-approach/target fallback; it does not silently select points1/2/3.

Mechanic route entries say Workstation Access Stops, not a misleading first walk to bay. Physical flow remains Break -> Customer Car (reverse Arrival walk) -> Service Bay (drive/park) -> Workstation (walk/repair) -> Drop-off (drive/park) -> Break (saved return walk). Existing arrival routes are not reordered. ON and OFF remain distinct; no occupied-car teleport fallback on unpaid arrival.

Parking and paid recovery

UpdateVehicleParking retries native parking ONCE after the existing parking timeout plus15 seconds. It logs direction/order/Paid for genuine retry/failure only. Persistent unpaid bay failure uses existing cancellation/accepted quote recovery, no premature repair payment. Persistent paid pickup failure enters session PaidReturnRecovery instead of canceling a good paid return. Approach/driver-loss and return enter/exit failures follow the same direction-aware policy.

FailVehicleTransfer logs ID, direction, Paid, stage, slot, elapsed, target/actual position, squared distance and speed when available. Original generic .5 messages cannot retrospectively prove direction. No every-tick log or additional world enumeration.

ProcessPaidReturnRecovery is eligible at most once per five GTA game seconds on the existing ~one-second visual cadence. Require a valid, stopped finite-speed car. Ask the assigned driver to leave at most twice, then HOLD if still inside. A later update must prove exit; another vehicle containing the worker is not touched. Validate empty driver seat/passengers, player not inside/near destination, destination vehicle/NPC clearance and existing return priority before the existing one-shot pickup path revalidates and moves the car. Unsafe conditions retain paid state; no extra settlement or invoice. No forced actor deletion.

ReleaseRecoveredMechanic frees the physical worker once after pickup placement and starts the existing return-to-break route. RecoveryWorkerReleased prevents replay. MechanicOnVehicleTask protects held recovery from competing employee animation/tasks. NeedsDropoffSoon keeps paid return ahead of new intake. Manual Vehicle Driving Options row7 resets bounded physical return attempts only; ignores unpaid/already-departing/normal OFF orders. OFF pickup continues its original automatic retry.

Ordinary OFF destination flags remain byte-identical after normalizing the conditional paid-recovery ped flag. Paid recovery explicitly checks NPCs as well as vehicles and the player. Moving/non-finite/negative speed never counts as stopped. A nearby actor can therefore HOLD recovery indefinitely until safe; no unsafe timeout cancellation is added.

Queue and status

A staged queue car gets one parking retry on the SAME entity, ID and FIFO reservation before original cleanup/retained-request behavior. The successful .5 three-customer queue, independent bay assignment and parts reservations remain protected. Repeated native parking cannot repair an undriveable MLO point; actual geometry needs owner correction.

MechanicServiceStatus differentiates startup/closed/paused/remote, worker unavailable, pickup/parking stage, paid return, vehicle absent/outside bay, workstation wait, valid repair and autonomous job with a quote. ShowEmployeeStatus and the explicit snapshot page provide slot/name/specialty/wages and route masks; recovery notices alone do not claim work. Work/car proximity warning is diagnostic only. No coordinates or progression change.

Phase 2 — financial/UI features

Owner capital transfers

Management has seven rows: Employees, Customers, Inventory, Accounting, Placement, Shop Location, Back. Owner transfers are solely under Accounting. Cycle session amounts $1k/$10k/$100k, open separate deposit/withdraw review, then Confirm. Merely viewing/selecting amounts/report period/back never moves money.

TransferPlayerMoney validates the selected amount, existing Story-ped restriction, wallet/business funds, wallet integer limit and long owner-counter/cash limits. Original Save/wallet rollback remains, including notification if an external mod prevents wallet restoration. Deposits/withdrawals remain capital, not revenue/expense. Selection state is not a new saved setting.

Retail sale draft and atomic batch

Parts Sales has four quantity rows, Review, Clear Draft and Back. Left/Right or Enter adjusts 0–100 per category; shared input arbitration/debounce remains. Review snapshots quantities and shows stock/unchanged unit-cost estimates/proceeds. Confirm rechecks CURRENT unreserved stock and ledger limits.

TrySellParts validates all lines, negative prices and overflow before mutation. One Save applies all stock/cash/gross retail/unit counters, with full rollback if the save fails. One-frame repeat guard remains. Successful CSV audit preserves the original six columns and emits one row per nonempty category, after commit only. Pending supplier cargo and promised repairs are never sold. No retail markup/margin change; zero estimate remains valid as before. Safe UI total formatting reports overflow instead of throwing while drawing. TrySellPart remains a one-unit wrapper for inherited entry-point invariants.

Employee/controller/menu help

Employee Status is explicitly a SNAPSHOT: three worker rows, receptionist, manager, Refresh/Full Status and Back. Native worker/car checks happen on open/refresh/detail selection; drawing reads cached status strings. Reception readiness requires a live worker; manager distinguishes remote simulation, absent stream and workstation route. Working is not inferred from HasJob/PhaseWorking alone.

ControllerSettings enum/page/action/navigation is removed; controller runtime mappings, debouncing, native pause and walk/look handling remain. ResolveControllerEnabled makes Settings.ini [Gamepad] Enabled authoritative when present; missing key falls back to legacy Save.ini/default. Existing save key is still written through unchanged Save, not deleted. No forced disable, new key or Settings.ini write from browsing.

Every highlighted row gets contextual text or the existing precise row help with the selected label. All67 pages and392 row footers compile/exercise through adapters; NAVIGATION_MATRIX.md and MENU_FOOTERS.md provide complete evidence. Back restores hub/role/editor/review parent and row. Existing wrapping/right-column fit/cache remain bounded; actual controller hardware and text truncation need GTA QA.

GTA-hour demand

AdvanceCustomerDemand adds only clamped sampled active gameplay seconds under boot/open/arrivals/not-paused/not-remote gates. It is the ONLY new managed call added to OnTick after existing economy processing; original clock sampling/order is source-pinned. QueueDueCustomer reads GTA GET_CLOCK_HOURS once on the existing visual cadence, not during constructor or every frame. No UTC deadline or offline catch-up loop.

Existing SecondsBetweenCustomers is the base (30–900): 00–05 x4; 06–08 x0.85; 09–17 x0.70; 18–20 x0.90; 21–23 x2, ceiling and bounded20–3600 active seconds. At base150: 600,128,105,135,300 seconds respectively. GTA hour weights REAL ACTIVE gameplay seconds; it is not a GTA-minute timer. Session elapsed resets across reload and shop/service transitions; new request creation only is weighted. Manual scheduling bypasses demand wait within capacity, and paused manual entry is rejected. No pending/accepted/paid request deletion, revenue or wage change.

Code, compatibility and preservation

Authoritative source/project remain experimental/shvdn3/Install/scripts/GlassHeroesBusiness.3.cs and existing directly linked net48/C#7.3 project. No new runtime dependency. Full source.diff against22bff09 is verified to reconstruct exact final source; SOURCE_SHA256.txt identifies it.

Preservation.json lists all changed/added methods and sole removed ExecuteControllerSelection. 332 existing methods remain byte-identical. Extra exact assertions freeze Save, FinanceV1 serializers/commit, TrySettleRepair, RecordCustomerInvoice, ProcessOneCustomerRepair, AssignCustomerMechanic, stock reservation, OnAborted, all point capture/load/save/render helpers, manager remote operations, ProcessEconomy, supplier simulation/receipt, cancellation and teleport retry clock. Constructor delta is only .6 BOOT label plus controller precedence; no early game native. OnTick delta normalizes exactly to the original after removing the managed demand call. Original OFF clearance is unchanged; only paid-recovery destination peds are additionally checked.

Save.ini/Settings.ini schema changes: NONE. Existing positions, headings, masks, staff XP, funds/journal/inventory/accepted quotes/supplier liabilities preserved. New recovery, demand and UI draft fields are session-only. Added CustomerStage value is appended and not serialized as a pending movement. Removed MenuPage value affects only unsaved UI state. Original v2 source/project hashes unchanged. .5 branch, PR32, main/QA and older worktrees remain protected.

Before: parking timeout canceled both directions using the same message. After: one retry -> unpaid existing promise recovery OR paid retained exit/empty-car recovery. Before: Owner Transfers hardcoded1000 immediately; after: amount + review + guarded Confirm. Before: one retail unit immediately; after: reviewed quantities/all-line commit. Before: stale Save.ini controller flag overrode config; after: explicit config wins, legacy fallback remains. Before: new requests used UTC deadline; after: sampled active elapsed plus GTA-hour weighting. No broad redesign.

Before / after code excerpts

Paid vs unpaid parking (full context in source.diff):

```
// Before: same cancellation for bay and paid pickup.
else if (order.SecondsInStage > _parkingTimeoutSeconds + 15)
    CancelCustomerOrder("Mechanic failed to park at saved point; revise heading", true);
// After: one bounded retry, then direction-specific handling.
else FailVehicleTransfer(order, atBay, "Parking failed at saved point; review
heading/clearance");
```

```
// Within FailVehicleTransfer:
if (order.Paid && !toBay)
{
    order.PaidReturnRecovery = true;
    order.Stage = CustomerStage.PaidReturnRecovery;
}
```

Route mask handling preserves the loop and data:

```
// Before: unset selected nodes are silently skipped.
if (_serviceRoute[i].IsSet && (routeMask & (1 << i)) != 0)
    result.Add(_serviceRoute[i].Position);
// After: warn once at route construction, then retain the same loop.
if (slot < _mechanics.Count) WarnEmptyWorkRoute(slot, false);
```

Controller precedence and owner amount:

```
// Before:
_controllerEnabled = data.GetValue("Gamepad", "Enabled", _controllerEnabled);
// After:
_controllerEnabled = ResolveControllerEnabled(_configuration, data, _controllerEnabled);
// Before wallet mutation used literal 1000. After validation:
int amount = _ownerTransferAmount;
if (amount != 1000 && amount != 10000 && amount != 100000) return;
```

Retail and demand boundaries:

```
// Before: single-unit entry mutated one bucket then saved.
// After: the one-unit entry delegates to the all-line transaction.
int[] quantities = new int[4]; quantities[kind + 1] = 1;
return TrySellParts(quantities);
// Before QueueDueCustomer compared now with a UTC deadline.
// After: only active elapsed can reach a GTA-hour-weighted request interval.
int hour = GTA.Native.Function.Call<int>(GTA.Native.Hash.GET_CLOCK_HOURS);
_customerDemandIntervalSeconds = CustomerDemandInterval(hour, _customerIntervalSeconds);
if (_customerDemandElapsed < _customerDemandIntervalSeconds) return;
_customerDemandElapsed = 0;
```

Status boundary:

```
// Before: PhaseWorking alone increased the working aggregate.
if (employee.ArrivalPhase == PhaseWorking) working++;
// After: row status requires live worker/car/assignment and actual stage gates.
_employeeStatusRows[i] = MechanicServiceStatus(i);
```

Actual tests and test boundaries

Full exact-source final run: actual Release0 errors/two inherited warnings (CS0169 unused _selectedInvoiceMechanicSlot, CS0649 CustomerOrder.VehicleTaskIssued). 762/762 behaviors: 85 original issue/logger/migration plus inherited feature/queue/financial tests, 180 navigation, recovery21, owner13, retail15, status15, demand13. Static21/21; scope/preservation PASS. All counts come from actual successful runs; earlier failed runs were fixture compile gaps, not silently counted passes.

Inherited migration/hardening adapters now include the actual managed demand helper; wage fixture adds a name for status rows. Single-sale fixtures include the real new batch core, input fixtures add the real retail quantity helper, and one old route-geometry fixture stubs its new warning side effect while focused recovery tests compile it verbatim. Obsolete controller/direct-transfer navigation cases were replaced with new confirmation/config cases, not treated as runtime UI coverage. Focused tests use actual extracted production members/case arms with deterministic native/entity/movement/settings adapters and temporary files. They do NOT run GTA navigation/parking, animation, hardware, NTFS permission semantics or Enhanced runtime.

Reproduce after SDK activation:

```
source /workspace/.cloud-setup/env.sh
python verification/workflow336/verify.py --label v0.3.3.6-final --profile workflow336 --suites
repair sales pallets wages reload supplier range menu cart pause focus finance startup hardening
 bays workflow335 recovery336 owner336 retail336 status336 demand336
```

Remaining limits and next gate

MLO paths, car/work geometry, driver native exit and animation must be verified in GTA. An unsafe or unresponsive occupied paid car holds after two exit attempts until manually retried/cleared; nearby target NPCs deliberately block it. No actor is forcibly deleted to manufacture safety. Unpaid bay failure remains cancellation/promise recovery after bounded retry, not unsafe occupied-car warp.

Paid cars and unaccepted roadside entities remain session transient after reload/abort, inherited from .5. Ledger remains paid; completed-car storage/persistence is NOT introduced. Destroyed/removed paid entities cannot be physically restored. Original best-effort CSV logging can fail independently of committed financial state; do not treat a missing CSV line as proof of a second payment.

Employee status is an explicit refresh snapshot, not a live overlay. Long footer text can truncate to the existing two-line footer; source/report contains full wording. Automatic demand first arrivals may wait longer than the former short startup deadline, especially overnight; manual scheduling is available for testing. Physical creation retry uses its existing separate active-game backoff.

No original private log/save was fabricated, no crash cause proven, and no .6 GTA result claimed. Owner should first validate P0 service/mask/return behavior on GH+LS RED, then UI/transactions/demand and incremental other mods. The checklist includes the reported mask8/unset4 and Bay3 layout suspicion without forcing a setup change. A source warning is not proof the native service workflow now succeeds in every MLO.

Review, rollback and package plan

See GTA_TEST_CHECKLIST.md for the detailed manual test sheet and complete paired-backup rollback. Keep .5 exact source and COMPLETE data/journals/backup files. Never alternate modified sole Save.ini between independent candidates. No install/rollback performed by Codex.

Prepare draft PR against feature/shvdn3-335-customer-repair-workflow, linking Issues33/34 and this guide, source diff, fresh logs, complete menu evidence and versioned reports. Preserve PR32 draft/unmerged. After source review, a separate authorization can request a single-source v0.3.3.6 RC1 archive; this milestone creates NO archive, tag or release and begins no next feature work.

Recorded automated results

Genuine SHVDN3 3.6.0 net48 / C#7.3 Release: PASS, 0 errors; inherited warnings CS0169 and CS0649. SDK 8.0.426.

762/762 behavior tests PASS: 85 inherited issue/logger/migration checks plus repair 32, sales 29, pallets 25, wages 22, reload 41, supplier 37, range 40, menu 13, cart 32, pause 9, focus 8, finance 33, startup 18, hardening 9, bays 34, workflow335 38, recovery336 21, owner336 13, retail336 15, status336 15, demand336 13; navigation 180. Static 21/21 PASS; focused scope/save/v2/clock/settlement/transfer-helper preservation PASS. The new navigation tests use production members/case arms with deterministic adapters, not GTA.

Candidate source SHA256: 7fff30978e2dc4c3235656f6ca6865506d5cea53387b7eb6d83654d71c8a9d6b. Full diff verified to reproduce this source from 22bff09.

GTA testing NOT RUN. Windows execution NOT RUN. Enhanced runtime compatibility NOT VERIFIED. Crash cause UNATTRIBUTED. No release archive, runtime deployment, save modification or merge was produced by this task.

Glass Heroes v0.3.3.6 — owner GTA acceptance — NOT RUN

Use Legacy 1.0.3889.0 / SHVDN Enhanced 3.9.0.6, GH + Los Santos RED first, then the owner's gradual additional-mod tests. Compilation is against official SHVDN3 3.6.0; new runtime compatibility is NOT VERIFIED.

Owner's crash-free .5 session is encouraging evidence for that source, not certification of .6 or resolution of Issue28.

Review, backup and future installation gate

- [ ] Review .6 source/diff and documentation before authorizing any RC package. This task creates no tag, release or live installation. Preserve exact working .5 RC1 source/hash AND COMPLETE matching scripts/GlassHeroesBusiness data outside scripts: Save.ini, Settings.ini, financial journals, CSV, backup/recovery files and logs. Use disposable QA copies; preserve the sole original.

- [ ] If a future test package is approved, enable exactly ONE Business Manager source/DLL, matching .6 source hash and complete data backup. Keep GTA/runtime/mod conditions fixed. Confirm .6 BOOT/source hash, one instance and default 30-second gameplay gate, no startup/save errors. No scripts/runtime DLL changes from this development task.

- [ ] Compare all existing cash/journal, inventory, supplier identities/liabilities, staff XP/assignments, accepted quotes, route masks, XYZ/headings before editing. No settings/default reset or new schema should occur.

P0 — configuration and route diagnosis FIRST

- [ ] Open Placement > Walking Routes > Mechanics. With the owner's reported mask 8 and ServiceWaypoint4 unset, inspect USE / UNSET, selected/effective/unset mask diagnostics and one-shot warning. Confirm masks/points are unchanged. Manually save the intended fourth physical stop OR choose appropriate already-saved stops for each mechanic after inspecting the route. Do not blindly enable all points or reorder an MLO path.

- [ ] Review pickup arrival geometry separately: reverse Arrival stops govern Break -> Customer Car walking. Work masks select FIVE shared workstation-access walking nodes, independently per worker/direction. Inspect the reported long/backtracking arrival legs in GTA; the code does not silently sort/rewrite them.

- [ ] Inspect Bay3's separate car/work positions. The reported ~0.39m separation is a potential obstruction, not a validated collision cause. If needed, on disposable data manually place the work point beside the vehicle with clear standing room; never rely on a car marker to prove drivable/animation geometry.

- [ ] One customer/one mechanic: Break -> Car walking, enter, drive to bay, exit, walk to work point, animation/repair, one payment/parts debit/invoice, drive to pickup, worker return to break and customer departure. Test CUSTOM then AUTO LOT, physical transfers ON and existing one-shot OFF.

- [ ] Two/three mechanics and three customers: first repairs while next checks in only when separate staffed bay/drop-off/reception is ready. Verify Car2/Car3 FIFO, stable roadside entities, no shared bay/worker or reserved-stock reuse. Missing Car2 must not allow a newer replacement ahead of surviving Car3.

- [ ] Temporarily block queue parking on disposable QA. Same staged car gets one bounded parking retry; if it still cannot park, request survives and FIFO rules remain. Do not expect a native parking retry to overcome an invalid MLO point.

P0 — paid vs unpaid failure and recovery

- [ ] Safely impede UNPAID drive/bay parking: one parking retry; persistent failure follows existing cancel/accepted-quote recovery with NO premature revenue or stock consumption. Verify failure log identifies order, slot, BAY direction, Paid=false, stage, target/actual position, distance/speed/elapsed.

- [ ] Complete a repair, then obstruct physical return/parking. One parking retry precedes PAID recovery; order remains paid and reserved, with no second payment/invoice/parts debit. Failure log must identify PICKUP/Paid=true. Prior .5 generic messages cannot retrospectively identify failure direction.

- [ ] Paid recovery asks the script-owned driver to exit, at most twice on bounded cadence. Confirm an empty car on a later update before any movement. Blocked/occupied/moving car, player inside or near target, passenger/driver still seated, NPC or unrelated vehicle at target must HOLD. No occupied-car teleport, forced removal of unrelated actors, repeated drive tasks or log spam.

- [ ] Clear obstructions and stay at least 3m from pickup. Empty/stopped car returns once through existing clearance/one-shot path; worker releases and follows existing return route. Check cash/stock, paid order ID and invoice count remain unchanged. Watch partial native/route failure where naturally reproducible; no repeated position write after successful placement.

- [ ] Vehicle Driving Options > Retry Held Paid Car Return explicitly resets bounded physical recovery attempts. It must ignore unpaid, already departing and normal OFF-mode orders. OFF pickup retries automatically when safe and retains its original safety checks.

- [ ] Missing/destroyed vehicle or worker on disposable QA: safe cleanup/hold and paid ledger retained; no ghost job credit. Paid transient cars cannot be restored after physical entity loss/reload. Preserve logs instead of manipulating the only save.

- [ ] Observe work station animation itself. Recovery notices do not prove working. New status should distinguish pickup walking, car driving/parking, workstation wait, actual repair, paid return and worker unavailable. Report exact stage/IDs and source hash for unresolved engine stalls.

Business/UI

- [ ] Management now has Employees, Customers, Inventory, Accounting, Placement, Shop Location, Back; direct cash shortcuts and controller menu are absent. Accounting > Owner Cash Transfers remains accessible.

- [ ] Cycle $1k/$10k/$100k. Review then confirm eligible Story Mode deposit/withdraw; insufficient funds/wallet limit/overflow/freemode ped cannot transact. Browsing/back/period toggle moves no money. Compare capital ledger versus revenue/expense. For save-failure testing use safe synthetic/test copies; verify wallet and business rollback, never corrupt a sole save.

- [ ] Parts Sales: four draft quantities, Left/Right and Enter adjustment, total preview, Review, Confirm and Clear Draft. Review/back/clear sells nothing. Confirm sells only current unreserved stock at unchanged unit-cost estimates, all lines in one save. Check gross proceeds/retail units and one compatible CSV row per nonempty category. Held key/controller echo must not sell twice; newly reserved stock must block stale drafts.

- [ ] Employee Status is explicitly a SNAPSHOT. Select to refresh worker rows and details; check bay/order/stage, selected/effective masks, wages, reception live readiness and manager physical/remote mode. Missing worker or stale PhaseWorking alone must not show repair. No continuous scans from holding this menu open; counts update on refresh.

- [ ] On disposable Settings.ini test explicit [Gamepad] Enabled=true with legacy Save.ini=false, and explicit false with legacy true. Settings wins. With setting missing, old save/default remains the fallback. Reload applies config. Controller input/debounce/native Esc/Start pause and custom mappings remain functional; no menu toggle writes Settings.ini.

- [ ] Use NAVIGATION_MATRIX.md for ALL 67 pages and MENU_FOOTERS.md for ALL 392 row descriptions. Test keyboard/controller accept/Back, wrapping/scroll, F2 close/reopen, role editors and exact return rows, stock/equipment/payments, point heading/load-save and native pause. Note clipping, stale snapshot or wrong action with exact selection path. Footers are bounded to existing two-line rendering; longer text can truncate.

Demand, regression and extended session

- [ ] With automatic arrivals enabled, compare equal ACTIVE durations around GTA 02:00, commute and daytime. Existing SecondsBetweenCustomers is the baseline; default 150s yields ~600 active seconds overnight, 128 morning, 105 workday, 135 evening, 300 late evening. These are real active gameplay seconds weighted by GTA hour, not GTA minutes or wall-clock deadlines; queue capacity still limits requests.

- [ ] Manual Schedule One bypasses automatic demand delay but respects capacity/open state. Changing GTA hour adjusts future request cadence only; pending/staged/accepted/paid jobs must not be removed. No demand while paused, closed, startup-held or in remote simulation. Restart uses fresh session interval; no offline catch-up.

- [ ] Regress manager offsite work/streaming, $5,000 income reconciliation with/without expense settlement, accrued versus paid bills, wages/XP, mixed supplier receipt exactly once, player-only wholesale, accepted quote/save

failure/reload, nine-box pallets and native pause controls. All remain independent from customer demand weighting.

- [ ] Reload/abort during intake, accepted repair and paid blocked pickup using COMPLETE paired backups. Unpaid accepted quotes retain identity/progress. Paid cars and unaccepted roadside requests remain session transient as in .5; no saved completed-car storage added. No additional payment after reload.

- [ ] Run the owner's chosen extended GH + LS RED observation, then incremental mod stack. Record active duration, menu paths, route mode/masks, order IDs/stages, actual animation and return direction. Preserve SHVDN/RPH/Errors logs and dumps. Do not combine CLR stack overflow, startup AV or EUP/file-access symptoms without evidence; no crash fix is claimed.

Rollback

Close GTA and preserve exact .6 source/hash plus full resulting data/logs separately. Restore the prior single .5 source and COMPLETE matching .5 business-data directory, including journals/recovery/backup/CSV files and Settings.ini. Do not restore only Save.ini or alternate independent builds against a modified sole save. A full rollback intentionally discards the test-session progress; keep the original backup untouched. Codex did not execute any live deployment/rollback.

v0.3.3.6 complete navigation matrix

All 67 production pages. Default Back uses neutral harness context; hub/role/editor/review parent overrides are separately tested. This is not automatic purchase/destructive-action testing or actual GTA hardware/rendering.

| Page | Title | Default Back / row | Items |

| --- | --- | --- | --- |

| Business | GLASS HEROES | Business / 0 | Open Business; Save Business; Business Status; Business Management > |

| BusinessManagement | BUSINESS MANAGEMENT | Business / 3 | Employees >; Customers >; Inventory & Suppliers >; Finances & Accounting >; Placement Settings >; Set Shop Location; < Back to Main |

| Employees | EMPLOYEES | BusinessManagement / 0 | Employee Status; Mechanics >; Receptionist >; Business Manager >; Employee Settings >; < Back to Business Management |

| EmployeeSettings | EMPLOYEE SETTINGS | Employees / 4 | Placement Settings >; Employee Walking Paths >; Blocked Path Recovery >; Placement Markers >; Break-Time Animations; < Back to Employees |

| EmployeePaths | EMPLOYEE WALKING PATHS | EmployeeSettings / 1 | Mechanic Walking Paths >; Receptionist Walking Paths >; Manager Walking Paths >; < Back |

| ArrivalWaypoints | SPAWN -> BREAK (0-5) | MechanicWalkingPaths / 2 | Arrival Point 1 / 5 >; Arrival Point 2 / 5 >; Arrival Point 3 / 5 >; Arrival Point 4 / 5 >; Arrival Point 5 / 5 >; < Back |

| MechanicBays | SERVICE BAYS | EmployeePlacements / 1 | Service Bay 1 >; Service Bay 2 >; Service Bay 3 >; < Back |

| ServiceWaypoints | SHARED WORK ACCESS (0-5) | MechanicWalkingPaths / 3 | Work Route 1 / 5 >; Work Route 2 / 5 >; Work Route 3 / 5 >; Work Route 4 / 5 >; Work Route 5 / 5 >; < Back |

| BayApproaches | BAY APPROACHES | MechanicWalkingPaths / 4 | Bay 1 Final Turn >; Bay 2 Final Turn >; Bay 3 Final Turn >; < Back |

| ServiceVehicleBays | SERVICE VEHICLE PARKING | CustomerPlacements / 6 | Service Bay 1 >; Service Bay 2 >; Service Bay 3 >; < Back |

| ServiceBayProfile | SERVICE BAY 1 | Business / 0 | Set Vehicle Position (CUSTOM); Set Mechanic Work Position; Mechanic Assignment; View Current Configuration; < Back to Service Bays |

| PointEditor | TEST POINT | Business / 0 | Set / Save Position + Facing; Clear Saved Position; Marker Preview; Show Saved Coordinates; < Back |

| RouteRecovery | ROUTE RECOVERY | EmployeeSettings / 2 | Blocked-Route Teleport; Recovery Status; < Back |

| PlacementMarkers | PLACEMENT MARKERS | EmployeeSettings / 3 | Placement Markers; Show Marker Color Legend; < Back |

| Customers | CUSTOMER SERVICE | BusinessManagement / 1 | Customer Order Status; Customer Queue >; Placement Settings >; Vehicle Driving Options >; Customer Arrivals; Schedule Next Customer; Customer Blips; Placement Markers; < Back to Business Management |

| Finances | FINANCES & ACCOUNTING | BusinessManagement / 3 | Financial Overview >; Income & Sales >; Expenses & Bills >; Profit & Loss >; Transaction History >; Owner Cash Transfers >; Reporting Period: Toggle; < Back to Management |

| FinancialOverview | FINANCIAL OVERVIEW | Finances / 0 | Reporting Period: Toggle; Period Closing Cash; Gross Revenue; Cash Flow Reconciliation; Outstanding Bills at Close; Save / Audit State; < Back to Accounting |

| IncomeSales | INCOME & SALES | Finances / 1 | Reporting Period: Toggle; Physical Customer Repairs; Manager Remote Repairs; Autonomous Mechanic Jobs; Parts Retail Sales; Gross Revenue (No Overlap); Known Rebates / Refunds; < Back to Accounting |

| ProfitLoss | PROFIT & LOSS | Finances / 3 | Reporting Period: Toggle; Gross Revenue; Cash Expense Payments; Expenses Incurred; Accounting Profit; Accounting Basis / Notes; < Back to Accounting |

| TransactionHistory | TRANSACTION HISTORY | Finances / 4 | #1 test history; Older Transactions >; Newer Transactions >; < Back to Accounting |

| OwnerTransfers | OWNER CASH TRANSFERS | Finances / 5 | Transfer Amount: Change; Review Owner Deposit >; Review Owner Withdrawal >; Recorded Owner Deposits; Recorded Owner Withdrawals; Reporting Period: Toggle; < Back to Accounting |

| StaffRoster | MECHANIC ROSTER | MechanicManagement / 0 | Mechanic 1 >; Mechanic 2 >; Mechanic 3 >; < Back to Mechanics |

| StaffProfile | MECHANIC PROFILE | StaffRoster / 0 | Profile / Performance; Training: +40 XP; Change Specialization; Raise Period Wage (+$50); Fire This Mechanic; < Back to Roster |

| VehicleDriveOptions | VEHICLE DRIVING | Customers / 3 | Drive-in Customer Cars; Mechanic Vehicle Transfers; Service Parking Mode; Auto Lot Settings >; Custom MLO Parking Bays >; Reverse Before Driving; Driving Status / Diagnostics; Retry Held Paid Car Return; < Back to Customers |

| CustomerPlacements | CUSTOMER PLACEMENTS | Customers / 2 | Customer Drop-off Parking >; Customer Road Spawn >; Customer Reception Approach >; Customer Check-in Position >; Waiting Car 2 Position >; Waiting Car 3 Position >; Service Vehicle Parking Bays >; Customer Walking Route (0-3) >; < Back |

| EmployeePlacements | EMPLOYEE PLACEMENTS | EmployeeSettings / 0 | Mechanic Break Area >; Service Bays >; Receptionist Desk >; Receptionist Desk Approach >; Manager Workstation >; Manager Workstation Approach >; Staff Spawn / Entry >; < Back |

| MechanicWalkingPaths | MECHANIC WALKING | EmployeePaths / 0 | Mechanic Walking Mode; Pickup + Return Exit Route; Spawn -> Break Waypoints >; Workstation Access Stops >; Final Bay Approaches >; Legacy Staff Checkpoint (MLO) >; Choose Job Waypoints by Mechanic >; Smooth Through-Waypoints; Teleport Stuck Pedestrians; Advanced Break <-> Bay Routes; < Back |

| ReceptionWalkingPaths | RECEPTION WALKING | EmployeePaths / 1 | Receptionist Walking Mode; Spawn -> Desk Waypoints >; Final Desk Approach >; < Back |

| ReceptionWaypoints | RECEPTION ROUTE (0-5) | ReceptionWalkingPaths / 1 | Reception Stop 1 / 5 >; Reception Stop 2 / 5 >; Reception Stop 3 / 5 >; Reception Stop 4 / 5 >; Reception Stop 5 / 5 >; < Back |

| CustomerQueueSettings | CUSTOMER QUEUE | Customers / 1 | Pending Customer Requests; Maximum Queue Capacity; Schedule One Customer; Clear Pending Requests; < Back to Customers |

| PartsSales | BUILD PARTS RETAIL SALE | InventoryManagement / 3 | Universal Parts: Quantity; Service Filters: Quantity; Brake Kits: Quantity; Diagnostic Supplies: Quantity; Review Sale >; Clear Sale Draft; < Back to Inventory |

| VehicleAutoLotOptions | AUTO LOT PARKING | VehicleDriveOptions / 3 | Car Parking Side; Car Offset from Mechanic Bay; Auto Parking Preview; < Back to Driving |

| BusinessExpenses | EXPENSES & BILLS | Finances / 2 | Outstanding Bills; Mechanic Wage Invoices >; Pay Receptionist Wages; Pay Rent & Utilities; Pay Supplier Invoices; Pay All Bills; Order 10 Universal Parts; Equipment & Office >; Financial Reports >; Pay Manager Wages; < Back to Business Management |

| MechanicInvoices | MECHANIC WAGES | BusinessExpenses / 1 | Mechanic 1 Wage Invoice; Mechanic 2 Wage Invoice; Mechanic 3 Wage Invoice; Pay Former Employee Wages; Pay All Mechanic Wages; < Back to Expenses |

| CustomerWalkingRoutes | CUSTOMER WALK (0-3) | CustomerPlacements / 7 | Customer Walking Stop 1 / 3 >; Customer Walking Stop 2 / 3 >; Customer Walking Stop 3 / 3 >; < Back |

| InventoryManagement | INVENTORY & SUPPLIERS | BusinessManagement / 2 | Inventory Overview >; Build Supplier Order >; Active Deliveries >; Parts Sales >; Stockroom & Pallets >; Supplier Settings >; < Back to Management |

| InventoryOverview | INVENTORY OVERVIEW | InventoryManagement / 0 | Universal Parts; Service Filters; Brake Kits; Diagnostic Supplies; Storage Capacity; Reorder Advice; Low-stock Threshold; < Back to Inventory |

| BuildSupplierOrder | BUILD SUPPLIER ORDER | InventoryManagement / 1 | Universal Parts; Service Filters; Brake Kits; Diagnostic Supplies; Supplier: Change; Cost / Capacity Preview; Place Purchase Order; Fill Low-stock Cart; Clear Cart; < Back to Inventory |

| ActiveDeliveries | ACTIVE DELIVERIES | InventoryManagement / 2 | Delivery Status; Order Contents; Recover Original Order; Cancel Pending Order; < Back to Inventory |

| EquipmentManagement | SHOP EQUIPMENT | BusinessExpenses / 7 | Equipment Status; Buy Tool Set; Buy Diagnostic Scanner; Buy Desk + Laptop; Place Tool Chest; Remove Tool Chest; Place Diagnostic Trolley; Remove Diagnostic Trolley; Place Desk in Front; Remove Placed Desk; Office PC Status; < Back |

| MechanicRouteRoster | MECHANIC JOB ROUTES | MechanicWalkingPaths / 6 | Mechanic 1 Job Route >; Mechanic 2 Job Route >; Mechanic 3 Job Route >; < Back to Mechanic Paths |

| MechanicRouteDirection | SELECT WALK DIRECTION | MechanicRouteRoster / 0 | Break -> Job Waypoint Selection >; Job -> Break Waypoint Selection >; Route Summary; < Back to Mechanic List |

| MechanicRoutePoints | JOB WALK WAYPOINTS | MechanicRouteDirection / 0 | Use Work Waypoint 1; Use Work Waypoint 2; Use Work Waypoint 3; Use Work Waypoint 4; Use Work Waypoint 5; All Waypoints ON; All Waypoints OFF; < Back to Route Direction |

| SupplierDeliverySetup | SUPPLIER SETTINGS | InventoryManagement / 5 | Physical Mule Delivery; Mule Road Spawn >; Truck Unloading Spot >; Driver Box Handoff >; Pallet Storage & Capacity >; Driver Walk Stops (0-3) >; Wholesale Warehouse: Fixed; Assign Current Pickup Vehicle; Assigned Vehicle / Cargo Status; Delivery Route / Stock Status; Recover Stuck Shipment; Cancel Pending Order (press twice); < Back to Inventory |

| SupplierDriverWaypoints | DRIVER WALK (0-3) | SupplierDeliverySetup / 5 | Driver Stop 1 / 3 >; Driver Stop 2 / 3 >; Driver Stop 3 / 3 >; < Back |

| HireMechanicRoster | HIRE A MECHANIC | MechanicManagement / 1 | Junior General Mechanic; Skilled Brake Technician; Experienced Diagnostic Technician; < Back to Mechanics |

| HireReceptionistRoster | HIRE RECEPTIONIST | ReceptionistManagement / 1 | Entry Receptionist (1 queue); Experienced Receptionist (2 queue); Senior Service Advisor (3 queue); < Back to Receptionist |

| MechanicManagement | MECHANIC MANAGEMENT | Employees / 1 | Mechanic Roster >; Hire Mechanic >; Mechanic Job Waypoints >; Mechanic Walking Paths >; < Back to Employees |

| ReceptionistManagement | RECEPTION MANAGEMENT | Employees / 2 | Receptionist Profile >; Hire Receptionist >; Receptionist Walking Paths >; < Back to Employees |

| ReceptionistProfile | RECEPTIONIST PROFILE | ReceptionistManagement / 0 | Profile & Queue Skill; Train Receptionist (+1 Capacity); Increase Period Wage (+$25); Dismiss Receptionist; < Back to Receptionists |

| SupplierPalletStorage | STOCKROOM & PALLETS | InventoryManagement / 4 | Storage Capacity Status; Purchase Additional Pallet ($750); Place / Move Pallet 1 >; Place / Move Pallet 2 >; Place / Move Pallet 3 >; Place / Move Pallet 4 >; < Back |

| ManagerManagement | BUSINESS MANAGER | Employees / 3 | Manager Profile >; Hire Business Manager >; Manager Walking Paths >; < Back to Employees |

| HireManagerRoster | MANAGEMENT CANDIDATES | ManagerManagement / 1 | Assistant Manager — 3% discount; Experienced Manager — 6% discount; General Manager — 10% discount; < Back to Management |

| ManagerProfile | MANAGER PROFILE | ManagerManagement / 0 | Profile & Business Reports; Train Manager (+40 XP); Pay Manager Wage Invoice; Dismiss Manager; < Back to Manager |

| ManagerWalkingPaths | MANAGER WALKING | EmployeePaths / 2 | Manager Walking Mode; Spawn -> Workstation Waypoints >; Final Workstation Approach >; < Back |

| ManagerWaypoints | MANAGER ROUTE (0-5) | ManagerWalkingPaths / 1 | Manager Stop 1 / 5 >; Manager Stop 2 / 5 >; Manager Stop 3 / 5 >; Manager Stop 4 / 5 >; Manager Stop 5 / 5 >; < Back |

| PlacementSettings | PLACEMENT SETTINGS | PlacementSettings / 0 | Customer Locations >; Employee Locations >; Service Bays >; Delivery & Stockroom >; Walking Routes & Approaches >; Marker Display Settings >; < Back to Management |

| DeliveryPlacements | DELIVERY & STOCKROOM LOCATIONS | DeliveryPlacements / 0 | Mule Road Spawn >; Truck Unloading Spot >; Driver Box Handoff >; Pallet Storage & Placement >; Supplier Driver Stops >; Equipment Placement & Upgrades >; < Back to Placement Settings |

| PlacementWalkingRoutes | WALKING ROUTES & APPROACHES | PlacementWalkingRoutes / 0 | Mechanics >; Receptionist >; Manager >; Customers >; Supplier Driver >; < Back to Placement Settings |

| MechanicPlacementRoutes | MECHANIC WALKING ROUTES | MechanicPlacementRoutes / 0 | Spawn -> Mechanic Break Area >; Workstation Access Stops >; Final Bay Approaches >; Job Waypoint Selection by Mechanic >; Legacy Mechanic Checkpoint >; Walking Mode & Recovery >; < Back to Roles |

| ReceptionPlacementRoutes | RECEPTIONIST WALKING ROUTES | ReceptionPlacementRoutes / 0 | Spawn -> Desk Stops >; Final Desk Approach >; Walking Mode >; < Back to Roles |

| ManagerPlacementRoutes | MANAGER WALKING ROUTES | ManagerPlacementRoutes / 0 | Spawn -> Workstation Stops >; Final Workstation Approach >; Walking Mode >; < Back to Roles |

| CustomerPlacementRoutes | CUSTOMER WALKING ROUTES | CustomerPlacementRoutes / 0 | Customer Walking Stops >; Reception Approach >; < Back to Roles |

| SupplierPlacementRoutes | SUPPLIER DRIVER WALKING ROUTES | SupplierPlacementRoutes / 0 | Supplier Driver Stops >; Driver Box Handoff >; < Back to Roles |

| OwnerTransferReview | CONFIRM OWNER WITHDRAWAL | OwnerTransfers / 2 | Review Amount / Business Cash; Confirm Transfer; < Back to Owner Transfers |

| PartsSaleReview | CONFIRM PARTS RETAIL SALE | PartsSales / 4 | Review Items / Proceeds; Confirm Sale; < Back to Parts Sales |

| EmployeeStatus | EMPLOYEE STATUS SNAPSHOT | Employees / 0 | Mechanic 1 (vacant); Mechanic 2 (vacant); Mechanic 3 (vacant); Receptionist; Manager; Refresh / Full Status; < Back to Employees |

Context returns: hub/role/editor retain caller/row; Owner review returns to deposit/withdraw row; Parts review returns to draft Review; Employee Status returns Employees row0. Management has seven rows and no cash/controller shortcuts. Physical paid recovery is Driving Options row7; OFF safe pickup remains automatic. Runtime controller input is configured in Settings.ini.